Agalvaan Field Manual

Run it, read it, fix it · October 2026

The three things that travel

Telemetry is the rover reporting on itself — battery, tilt, wheel speed. **Video** is two camera streams. **Commands** are typed into the rover's terminal over SSH.

Mission Control **never publishes to the rover**. It listens, and it types commands into a shell. That is deliberate: nothing the operator does can write into the rover's own message traffic.

*Both machines have **fixed addresses** (.50 and .20). The arena Wi-Fi name can change without anything else changing — see page 5.*

Starting up, in order

Each step needs the one before it.

1 Power the arena router FIRST

If the rover boots with no router, it cannot find the network and will sit on the wrong one.

2 Power the rover, wait about a minute

Teleop, battery logging and the rear camera all start by themselves.

3 Open the laptop; it joins automatically

Check with **ip -br addr** — you want **192.168.2.50**.

4 Plug the joystick into the LAPTOP, then start it

ros2 run joy joy_node — the rover reads it over the network.

5 Start Mission Control

./run_ui.sh — never **python3 -m mission_control.main**, which silently shows FAKE data.

6 Start the front camera by hand

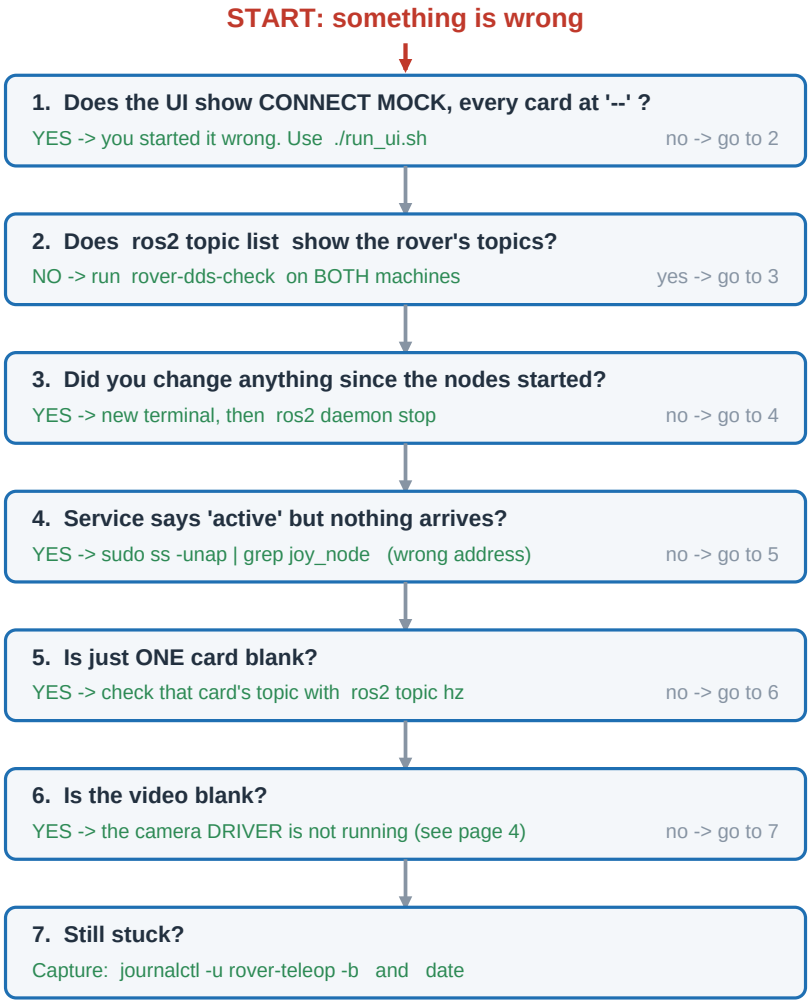
ros2 launch rover_bringup camera.launch.py on the rover. The rear camera is already running.

Two minutes of checks before a run

rover-dds-check on both machines · **date** on the rover (it must not say 1970) · the POWER card shows volts · the joystick moves the wheels.

When something is wrong

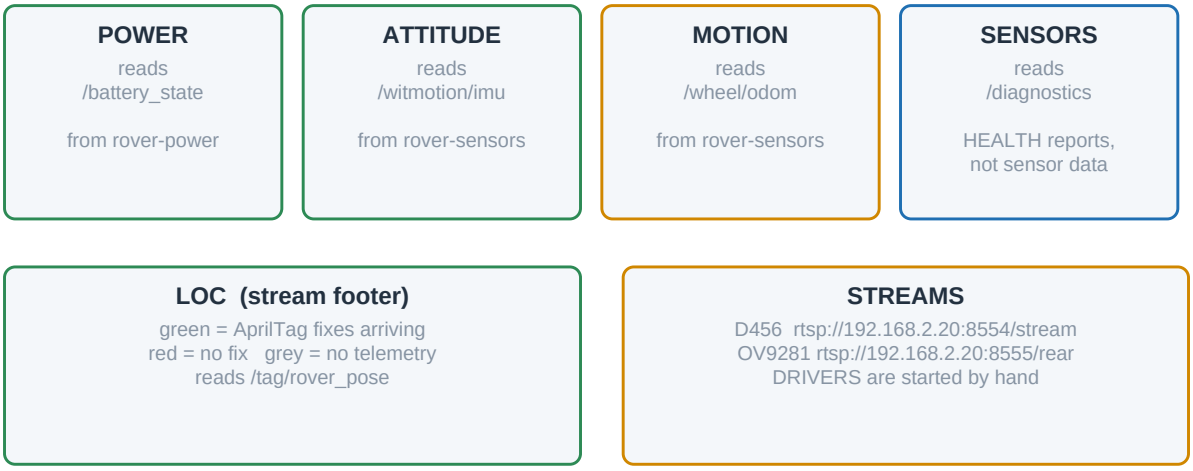
Start at the top. Do not skip ahead — step 1 is the most common fault by a wide margin.



Work DOWN the list. Each step rules out the ones below it.
This order matches what has actually gone wrong on this rover.

What each part of the screen reads

A blank card is not a broken card. It means its topic is silent, which usually means the thing that publishes it is not running.



The one that confuses people

SENSORS reports health, not data. The IMU can be publishing perfectly while SENSORS says nothing about it, because that card reads health reports and the rover does not publish one for the IMU. ATTITUDE showing numbers is the proof the IMU is fine.

If the rover explicitly reports a sensor as DEGRADED, that always wins over 'messages are arriving'. The rover knowing something is wrong beats us guessing from traffic.

Symptom to fix, at a glance

Symptom	Cause	Fix
Every card '--', CONNECT MOCK shown	UI started the wrong way	./run_ui.sh
ros2 topic list is empty	RMW / domain / URI mismatch	rover-dds-check on BOTH
Topics listed but no data	stale ros2 daemon	ros2 daemon stop
Service 'active', nothing arrives	node bound the OLD address	restart that service
MOTION blank, ATTITUDE fine	encoder ESP sending nothing	power the motor electronics
SENSORS says nothing for IMU	no IMU health report exists	normal — check ATTITUDE
Video blank	camera driver not running	launch camera.launch.py
Rover clock says 1970	no RTC battery	rover-clock restores it
Changed something, no effect	process read config at START	restart that process

If the arena Wi-Fi name is not Team_1

Two commands, one on each machine. Nothing else changes.

```
sudo nmcli con modify Team_1 802-11-wireless.ssid "Team_7" \
    wifi-sec.psk "<new-password>"
sudo nmcli con up Team_1
```

Everything else is keyed on the two fixed addresses, and those are **ours**, not the router's. The two machines keep talking even if the router hands out a completely different range.

The one thing that would really break it is being forced onto automatic addresses. Then the addresses move and the rover and laptop lose each other. Keep both ends fixed.

Commands worth knowing

Can this terminal talk to the rover?
```
rover-dds-check
```

Start Mission Control
```
./run_ui.sh
```

After changing ANY setting
```
ros2 daemon stop   + open a new terminal
```

What is the rover publishing?
```
ros2 topic list
```

Is a topic actually flowing?
```
ros2 topic hz /witmotion/imu
```

Which address did a node bind?
```
sudo ss -unap | grep joy_node
```

Are the video streams being served?
```
sudo ss -tlnp | grep -E ':8554|:8555'
```

Start the front camera
```
ros2 launch rover_bringup camera.launch.py
```

Service logs, this boot only
```
journalctl -u rover-teleop -b
```

Rover services at a glance
```
systemctl status rover-teleop rover-power rover-video
```

One rule that explains most surprises

Programs read their settings once, when they start. Changing a file or an address does nothing to something already running. After any change: restart the thing, and open a fresh terminal.